NEON FLASHING SIGNS: AI, SENSEMAKING, AND THE THRESHOLD OF THE PUBLIC LIBRARY MAKERSPACE

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A woman stands in the doorway of a public library makerspace. In front of her sit a 3D printer, a laser cutter, a vinyl press, and a wall of tools she cannot name. No sign tells her what the room is, who it is for, or whether she is allowed to touch anything. She reads the silence as an answer, and she leaves. In a recent study of four Kentucky library makerspaces, one participant summed up the institution's tacit stance toward people who struggled to understand what the makerspace was: "What, do they need neon flashing signs? Figure it out!"

They do, in fact, need neon flashing signs. Perhaps not literally, but the joke names a real failure. Across the makerspaces we have studied, the tools work fine. What breaks is sensemaking: the slow, effortful work of figuring out what a space affords, whether it affords anything for *you*, and how to begin, which public libraries typically offload as labor for the potential user to undertake. This paper argues that generative AI might help carry that load, and that the same body of critical theory which explains why makerspaces fail should govern how, and whether, we let it.

The Problem

Makerspaces have swept the public library world, from the first in Fayetteville, New York, in 2011 to roughly 30% of U.S. libraries by 2024 (2023 Public Library Technology Survey, 2024). They are pitched as engines of "21st century skills" (Adler-Beléndez et al., 2021; Bowler, 2014; Koh & Abbas, 2015). Yet many sit underused, and some are quietly rebooted or closed as failures. The reflexive explanation tends to blame the community: people are not ready, not interested, not skilled enough. Our work has consistently found the opposite. The barrier is rarely the equipment or the people. It is the invisible film of assumed knowledge stretched across the threshold.

Three findings recur through multiple datasets over more than a decade (Chugh & Crawford Barniskis, forthcoming; Crawford Barniskis, 2014, 2015, 2016a, 2016b, 2022). First, affordances are hidden or de-contextualized, so sensemaking becomes a heavy cognitive and social load. James Gibson (1977) called affordances what an environment "offers the animal ... for good or ill" (p. 68). A laser cutter affords nothing to someone who does not know what it is, cannot tell whether using it costs money, and has no one to ask. Little signage, almost no community outreach, and weak connection to the rest of the library leave the potential maker to intuit a whole world of possibility from an often-unlabeled machine. Elaine Gurian (2005) calls the hesitation that many people display when confronted by a makerspace boundary *threshold fear*, invoking the need for neutral ground from which to test whether to enter. A closed door, or a wall of unexplained tools, gives no such ground. A room filled with mysterious and intimidating processes will send all but the most confident and curious away. Second, many people want these spaces for social engagement, play, and regeneration more than for producing a specific object, the "casual leisure" that library rhetoric tends to treat as a lesser rung on the ladder to "serious" making (Crawford Barniskis, 2023). Third, librarian *habitus* (Bourdieu's (1977) concept of the durable, often-unexamined dispositions and worldviews) shapes these spaces profoundly, encoding assumptions about what counts as "making" and who the space is for. Making itself is culturally coded as white, male, and technical. The person in the doorway is often reading, correctly, that the room was not imagined with her in mind.

The sensemaking problem is not merely informational. It is a problem of power involving who gets to understand a space well enough to use it, and who is left to figure it out or leave. In Library and

Information Science this is sometimes described as a problem of sociocultural access, where people do not understand not just HOW to use a tool, but why they should bother or how it fits with the rest of their lives (Burnett et al., 2008; Mathiesen, 2014; Oltmann, 2009), and which, due to what Elfreda Chatman (1991) characterizes as a "small world" viewpoint, can be challenging to impact.

AI Potentials

Generative AI is being offered to galleries, libraries, archives, and museums as a solution in search of problems. Sensemaking at the makerspace threshold is a real problem that AI might actually fit. Consider the failures above, reframed as questions a well-designed system could help answer for a specific person, in their own language, at the moment they are hesitating in the doorway: *What is this space? What could someone like me make here? What would it cost, how long would it take, and where do I begin?* An interactive system that could take a half-formed wish ("I want to make something for my granddaughter") and translate it into a legible path through the room's actual affordances would be doing sensemaking work that signage alone cannot. It could narrate the space in multiple framings, for multiple lives, in a way that a single static label never will.

RQ1 for this workshop asks: "How can human-AI interaction enrich, and be enriched by, the domains of digital humanities and cultural heritage?" In the library makerspace context, human–AI interaction could enrich cultural heritage practice by easing the threshold, We propose using AI to assist unfolding the messy, situated experiences in makerspaces, shaking open the seams so people can see if the garment will fit them. To build this AI tool, participatory design workshops could add to the knowledge we have already garnered through ethnography, memory maps, repertory grids, traffic sweeps and other research methods we have used, which all identify the gaps between what a library says it offers and what people perceive. Such participatory design could in turn stimulate new interaction methods for surfacing tacit affordances to address this workshop's RQ2 (How can the challenges and datasets inherent in these domains stimulate innovation in human-AI interaction methodologies and frameworks?). We have begun this work. With participatory design workshops at libraries creating new makerspaces and through research leveraging each maker's *imagined map* of a space they wish existed, we are attempting to build solutions for cash and resource-strapped public libraries. The participatory design processes are oreinted to create design specifications that current recommender paradigms have no idea what to do with, if such recommender paradigms even exist in libraries. Often they do not. Many libraries are offering makerspaces without even minimal staffing assistance, or signage.

However, our own use of theory makes us wary of this proposal to use AI, and the wariness *is* the argument. Ivan Illich distinguished *convivial tools*, which forward power to their users to decide how, why, and when to use them, rather tools that engineer dependence. Such tools that produce, in Illich's words, a "radical monopoly" (Illich, 1973) where the institutional way becomes the only thinkable way. Libraries are struggling to build such convivial tools, and are often defaulting to less convivial institutionally-mandated experiences (Crawford Barniskis, 2016a, 2016b, 2022, 2023). An AI sensemaking layer could go either way, convivial or prescriptive. Designed convivially, An AI interface could offer the newcomer enough orientation to make her own choices and then get out of the way. Designed carelessly, it could become one more institutional voice telling people what making is *supposed* to look like, automating librarian habitus at scale, hiding its assumptions inside a friendly interface, and calling the output "empowerment." That is a *conviviality mask* (Caire et al., 2011; Caire et al., 2008): institutional control wearing a false face of user agency. A toddler offered three pre-approved choices for supper is not thereby empowered; they are given pre-selected palatable choices, but actual choice is not on the menu. Similarly, potential users of makerspaces would not be empowered by pre-selected choices made by library staff and embedded in an AI interface. Institutional-centered design would reinforce and tighten organizational control on how users might benefit from (i.e. have access to: Ribot & Peluso, 2003) the makerspace.

Guardrails for a Liberatory AI System

If the question is what guardrails these systems need and who decides, the answer cannot be handed to the vendor whose incentive is engagement. It should be governed by the values libraries already claim. Part of these commitments entail what Crawford Barniskis (2025) calls IDEAL: Inclusion, Diversity, Equity, and Access, (for the purposes of Liberation and/or Love). The purposes matter; they are the foundational reasons for what is commonly termed DEI to matter. In this case, four commitments are non-negotiable.

Forward power; do not model it. The system's job is to widen the newcomer's sense of what is possible, not to narrow it to what the model predicts she wants. It should offer plural framings. The space as a place to build skills, yes, but also to play, to gather, to rest, to make a gift—because the casual-leisure user is exactly the person current "uplift" narratives already fail (Crawford Barniskis, 2023). A system trained to optimize toward "serious" making would concretize an existing exclusion into code.

Make the assumptions inspectable. Habitus does its damage precisely by staying unexamined. Any system mediating who understands a space owes its users, and its librarians, a legible account of why it surfaced *this* suggestion and not another. A black box that cannot show its reasoning cannot be audited for the biases the research on library makerspaces keeps finding (e.g. Higginbotham, 2020; Mann, 2020; Melo, 2020; Melo & Rodney, 2023; Sanchez, 2020).

Protect the people in the doorway. The population that most needs sensemaking help is also the most vulnerable. These are newcomers, people in socioeconomic precarity, those for whom the library is one of few remaining third places, and as our allied work on trauma-informed design makes clear (Chugh & Crawford Barniskis, forthcoming), people carrying trauma into a space that should feel safe. A sensemaking system that logs a hesitant newcomer's every half-formed question, or that ports her stated desires into an ad-driven recommender betrays the very trust that makes the threshold crossable. Privacy here is not a compliance checkbox, it is a precondition of threshold safety.

Keep the human enzyme. Elsewhere we have described the librarian's "enzymatic" role (Crawford Barniskis, 2016a). Librarians act similar to enzymes in a compost heap, getting things heating up, connected, generative. In a makerspace they do this by actively engaging with newcomers and breaking down barriers between people and possibilities. Such librarians help a newcomer feel welcome in these often-intimidating spaces. However, many libraries cannot afford a constantly staffed makerspace, and the disposition to be enzymatic is not universal. The AI should scaffold that enzymatic librarian role, not replace it. A machine can narrate a room; it cannot yet notice that someone is about to give up and walk over to say *you're welcome here*. But it could help narrow sensemaking gaps when staff are not present.

Envisioning an Ethical AI Solution

We envision a tablet or touchscreen-based AI system that could be posted at the door of a makerspace. This system should, in our view, be co-designed with community members to answer the questions they have in meaningful ways while protecting their privacy. This system could broadcast flyers, perhaps of upcoming events, until interacted with, then shift to the goals or questions that a user might have. A person at the threshold could use it to discover what tools are available, what has been made in the past, what might be possible, and what the rules of engagement are. There could be short videos of other makers doing show and tell of a project or introducing their expertise if they wish to collaborate or assist people. There could be user check-in opportunities, to allow those who are seeking social interaction to request it (Bilandzic & Foth, 2014; Bilandzic & Johnson, 2013; Bilandzic et al., 2013). It should be programmed to be actively welcoming and embody a Derridian radical hospitality (i.e. less conditional or bounded by library culture than is the hospitality standard in many of these spaces, Derrida, 2005) to assist people past any threshold anxiety, and to avert epistemic ignorance of the possibilities of these spaces across different ways of knowing or being in the world (Kuokkanen, 2008). We invite feedback to help us begin the prototyping process, but will be developing solutions with local libraries and community members as well. Participatory service design is the best methodology for this development (Holmlid, 2009), as it takes into account the interlocking aspects of the physical space, staff and user interactions, signage, tools, policies, rules, and so on.

We encourage the system to focus on organizing structures of the space's use, which we have defined through the following five subsystems: Welcoming and Belonging, Resistance and Recovery, Learning, Creative Expression, and Production. We identify these systems based on interviews, ethnographic observation and participation, repertory grids, visual traffic sweeps, imaginative mapping exercises, and participatory design activities, and they encapsulate the diverse goals of the people using the library's makerspaces. The first four of these subsystems is bounded by the fifth, because production is also a core activity within each of the subsystems, but some people may be interested solely in production and not the other subsystems. These subsystems are all part of the system of the makerspace, which is itself a subsystem of the library. A potential framing for the interface of the AI could involve these subsystems and supporting sensemaking around each one, answering potential questions such as: what is this goal about, how does this subsystem function, what are the possibilities in light of this subsystem? By focusing on the goal-centered subsystems, the interface could surface needs and desires of which potential users might not be previously aware. This opens the makerspace to alternate uses and expands the user's view.

A Conclusion, and a Caution

The makerspace failures we have documented are often sensemaking failures dressed up as community deficits, and obscuring prescriptive librarian habitus. That reframing is what makes AI worth taking seriously in this case, not as a flashier tool for the wall, but as a way to lower a threshold that libraries have left needlessly high. The same reframing is what makes us insist on the guardrails. A convivial sensemaking system, governed by library values and built through participatory, service-design methods *with* the communities it serves, could help the woman in the doorway understand that the room is hers. A nonconvivial one would give her a neon sign that flashes, in the end, the same old message about who these spaces are really for.

Our position is not "AI will fix makerspaces." It is narrower and, we hope, more useful. Sensemaking is the real point of failure. with AI as a plausible and specific intervention. The critical traditions that explain the failure—conviviality masks, habitus, threshold fear, the two-audience problem—are the only partially adequate guides to building the intervention without reproducing the harm. The rest of the sensemaking work must emerge from community members through participatory service design. They will design the enabling conditions, forward the power, and let people make sense of the space on their own terms. Otherwise we will have automated the shrug, adding merely a fancy AI hat: *Figure it out.*

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